

Customer Retention & Cohort Analysis

A cohort-based study of customer behaviour, retention patterns, and revenue dynamics on the Olist Brazilian e-commerce platform

Dataset Olist Brazilian E-Commerce Public Dataset (Kaggle)

Tools Google BigQuery (SQL) · Python 3 (Google Colab)

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Executive Summary

Across 93,357 customer accounts and over 96,000 delivered orders on the Olist Brazilian e-commerce platform (October 2016 – August 2018), this analysis reveals a severe retention crisis: **96.4% of customers never return after their first purchase**. Addressing this structural churn problem, without acquiring a single new customer, represents the single largest untapped revenue opportunity available to the business.

Key findings:

- Month-1 retention averages just **0.47%** across all cohorts — meaning fewer than 1 in 200 customers makes a second purchase in the month immediately following their first.
- Repeat customers are only **3.0%** of the customer base yet generate **5.6% of total revenue** (R\$ 864,357 of R\$ 15.4 million), with an average order value of R\$ 145.98 vs R\$ 160.76 for one-time buyers — indicating retained customers transact more frequently even if at slightly lower individual order values.
- Among the 2,801 customers who did return, the median gap to a second purchase was just **29 days** — defining a precise, actionable re-engagement window.
- The Platinum customer tier (top 25% by lifetime spend) accounts for **59.4% of total revenue** with an average spend of R\$ 392.68 — nearly 9× the Bronze tier average of R\$ 43.59.
- Credit card dominates at **75.3%** of all orders; boleto (cash payment slip) at **19.5%** signals a significant unbanked or cash-preference customer segment.
- A logistic regression churn model (AUC = 0.690) confirms that **spend quartile and month-2 retention are the strongest protective factors** against churn, while later acquisition cohorts carry statistically higher churn risk — validating the acquisition quality recommendation with predictive evidence.
- State-level geographic analysis reveals that **Rondônia (RO) leads all states at 4.3% repeat rate** — 43% above the 3.0% national average — suggesting measurable geographic variation in retention quality that can inform targeted acquisition strategy.

Three targeted recommendations — a timed re-engagement campaign, a Platinum customer retention programme, and a shift toward quality-focused acquisition — are presented with specific, implementable actions on the final page. A logistic regression model and state-level geographic analysis provide additional predictive and strategic depth.

Business Context

The retention problem in e-commerce

Customer acquisition is expensive. For any e-commerce business, sustainable profit lies in repeat purchases — yet most platforms invest heavily in acquiring customers who buy once and disappear. This analysis examines whether Olist faces this problem, how severe it is across cohorts and seasons, and where the highest-value customers originate.

What is cohort analysis?

A cohort groups customers by the month of their first purchase. Tracking each cohort over time reveals whether newer customers behave differently from older ones, and whether retention is improving, worsening, or structurally flat. It transforms a vague question — "what percentage of customers are active?" — into a precise one: "what percentage of customers who first bought in January 2017 were still purchasing in April 2017?" The latter drives decisions; the former does not.

Scope of this analysis

This analysis covers only delivered orders across 22 active acquisition months. It focuses on customer-level behaviour — retention, churn, segmentation, and repurchase timing — rather than product, seller, or logistics performance.

Dataset & Assumptions

Dataset overview

Attribute	Detail
Source	Olist Brazilian E-Commerce Public Dataset (Kaggle)
Date range	October 2016 – August 2018
Total customers	93,357 unique customer accounts
Delivered orders	~96,478 (used in all analysis)
Repeat customers	2,801 (3.0% of customer base)
Total revenue	R\$ 15,422,461.77
Tables used	orders, customers, order_items, payments (4 of 9)
Tools	Google BigQuery (SQL), Python 3 (Google Colab — pandas, seaborn, matplotlib)

Key analytical assumptions

- **Only "delivered" orders included.** Cancelled, processing, and shipped orders are incomplete transactions. Including them would inflate order counts and distort revenue and retention figures.
- **customer_unique_id used, not customer_id.** The dataset assigns a new customer_id per order; customer_unique_id is the true persistent identifier required for any repeat-purchase analysis.
- **Cohorts with fewer than 50 customers excluded from retention charts.** Small cohorts produce statistically noisy retention percentages that mislead more than they inform.
- **Revenue = sum of payment_value per order.** Includes freight costs, reflecting the true transaction value from the business perspective.
- **Churn defined as zero repeat purchases in the dataset window.** No arbitrary time-window definition is applied. Any customer with only one order across the full 22-month observation period is treated as not retained.

Analysis — Retention

Cohort retention heatmap

The heatmap below maps every cohort (rows) against months since first purchase (columns). Each cell shows the percentage of that cohort's customers who made a purchase in that month. Month 0 is always 100% by definition.

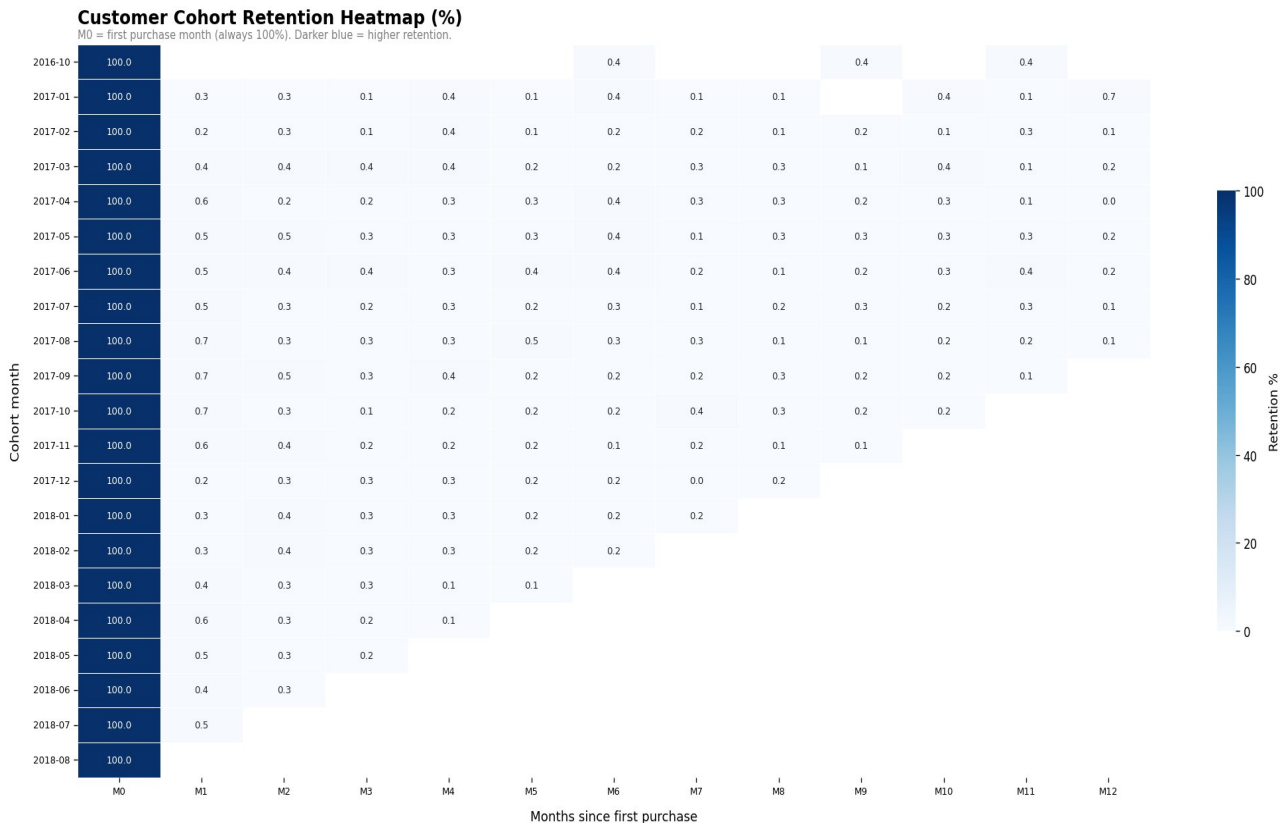


Figure 1: Cohort retention heatmap — % of each cohort active per month since first purchase.

Month-1 retention averages 0.47% across all cohorts — meaning that by the very next month after their first purchase, fewer than 1 in 200 customers returns. By Month 3, this stabilises at approximately 0.25%. The consistency of this collapse across every cohort — from October 2016 to August 2018 — confirms this is a structural behavioural pattern, not a one-off event. Olist customers are, overwhelmingly, one-purchase shoppers regardless of when they joined.

Retention curves by cohort

Five representative cohorts are plotted below to show the shape of retention decay over time and whether certain cohorts outperform others.

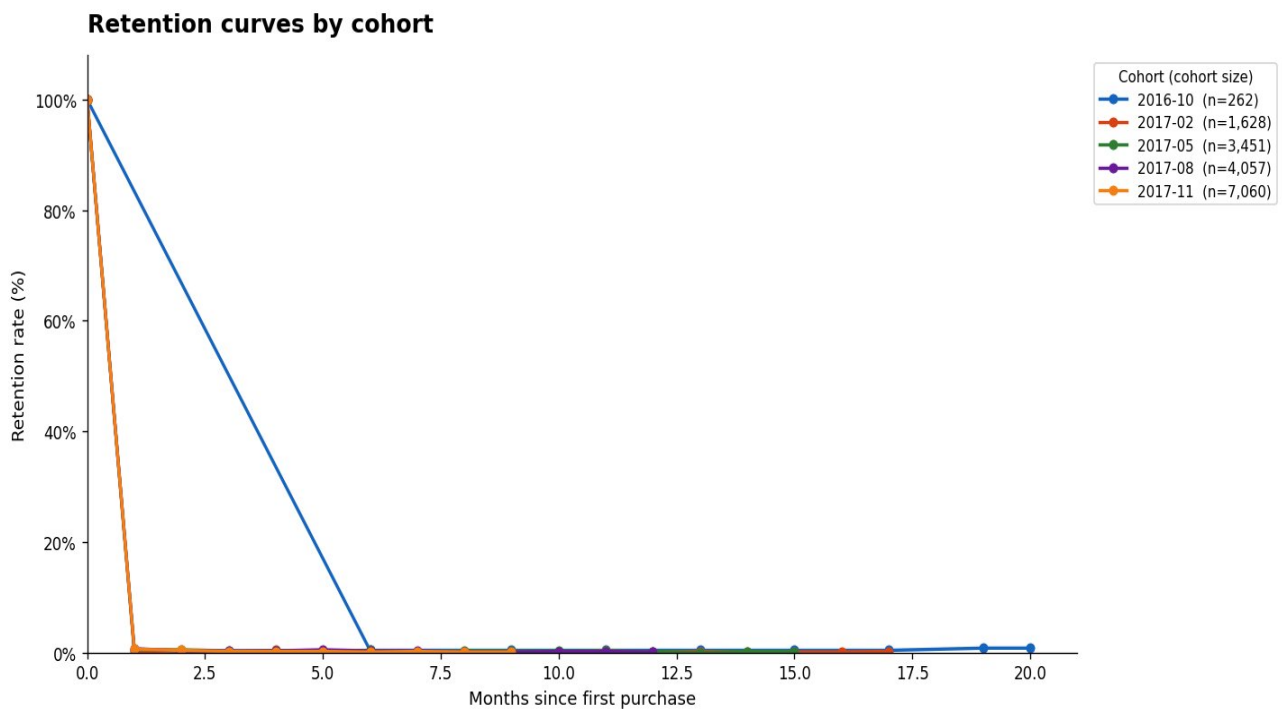


Figure 2: Month-by-month retention curves for five selected cohorts (cohort size shown in legend).

The August 2017 cohort shows the highest Month-1 retention at 0.7%, while the March 2017 cohort leads at Month 3 with 0.4%. After Month 4, all cohort curves converge near zero, suggesting a stable but very small loyal customer floor. There is no evidence of improving retention trends across cohorts over the 22-month period — the curves look structurally identical regardless of acquisition vintage.

Analysis — Churn & Acquisition

One-time vs repeat buyer rate by cohort

The chart below shows, for each cohort, what proportion of customers bought only once (churned) versus returned for at least one more purchase.

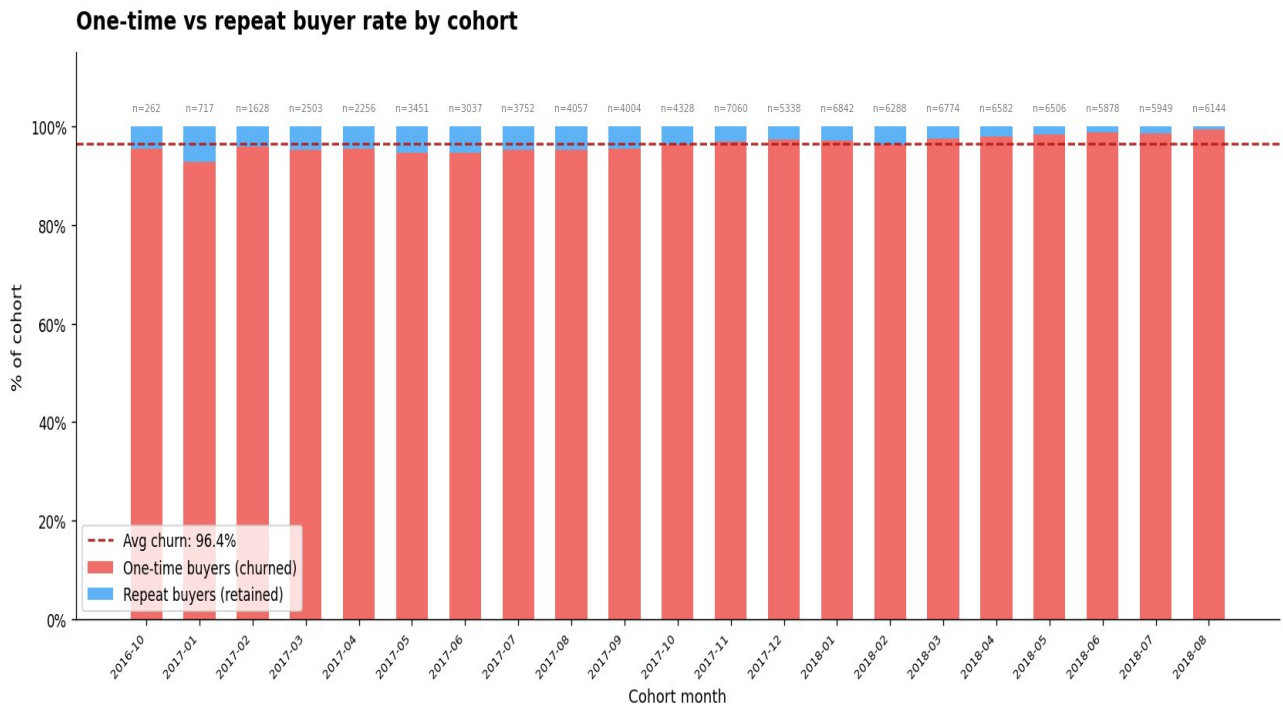


Figure 3: Stacked bar — churned (one-time) vs retained (repeat) customers per cohort, with average churn line.

The average churn rate across all valid cohorts is 96.4% — meaning 96 in every 100 customers never return. Notably, the October 2016 cohort (the earliest) shows a churn rate of 95.4%, marginally better than many later cohorts, possibly reflecting stronger early-adopter brand affinity. More concerning is the trend in 2018: churn rates reach 98–99% in the final months, though this is partly explained by insufficient time for those cohorts to generate repeat purchases before the dataset ends.

Monthly new customer acquisition

The dual-axis chart below shows new customer intake per month (bars, left axis) alongside cumulative customer growth (line, right axis).

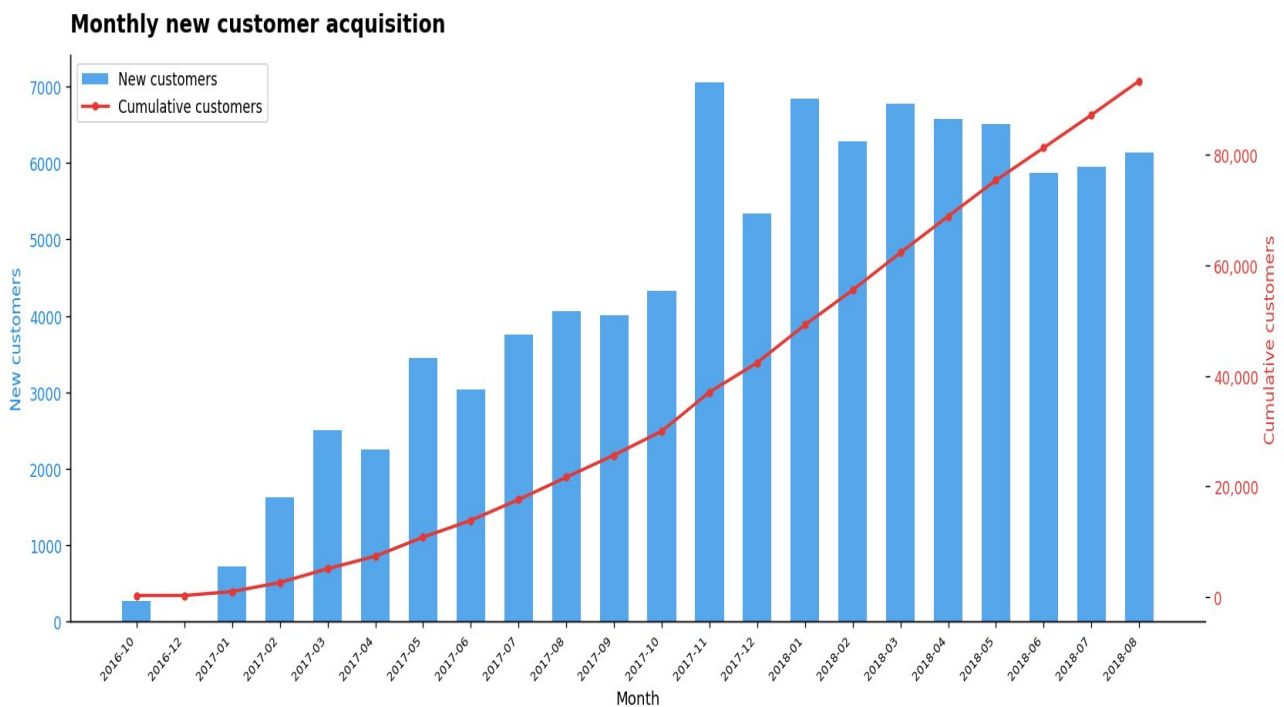


Figure 4: Monthly new customer acquisition (bars) and cumulative customer base growth (line).

New customer acquisition peaks in November 2017 at 7,060 customers — consistent with Brazilian Black Friday and holiday shopping behaviour. The cumulative base reaches 93,357 customers by August 2018. However, since Month-1 retention has not improved across this period, growing acquisition volume is not translating into a growing loyal customer base. Each acquired customer is, with 96.4% probability, a one-time transaction.

Logistic regression churn predictor

A logistic regression model was trained on cohort-level features to predict whether a customer cohort would exhibit high or low churn. Features were engineered from the existing KPI outputs: average order value, revenue per customer, cohort size, acquisition month index, payment method mix, month-1 and month-2 retention rates, and seasonal flags (holiday and mid-year). The dataset was expanded to customer-level rows — one per customer — using each cohort's known one-time and repeat buyer counts as ground truth labels. The model was evaluated using 5-fold stratified cross-validation.

Logistic Regression — Churn Prediction | Olist Cohort Analysis

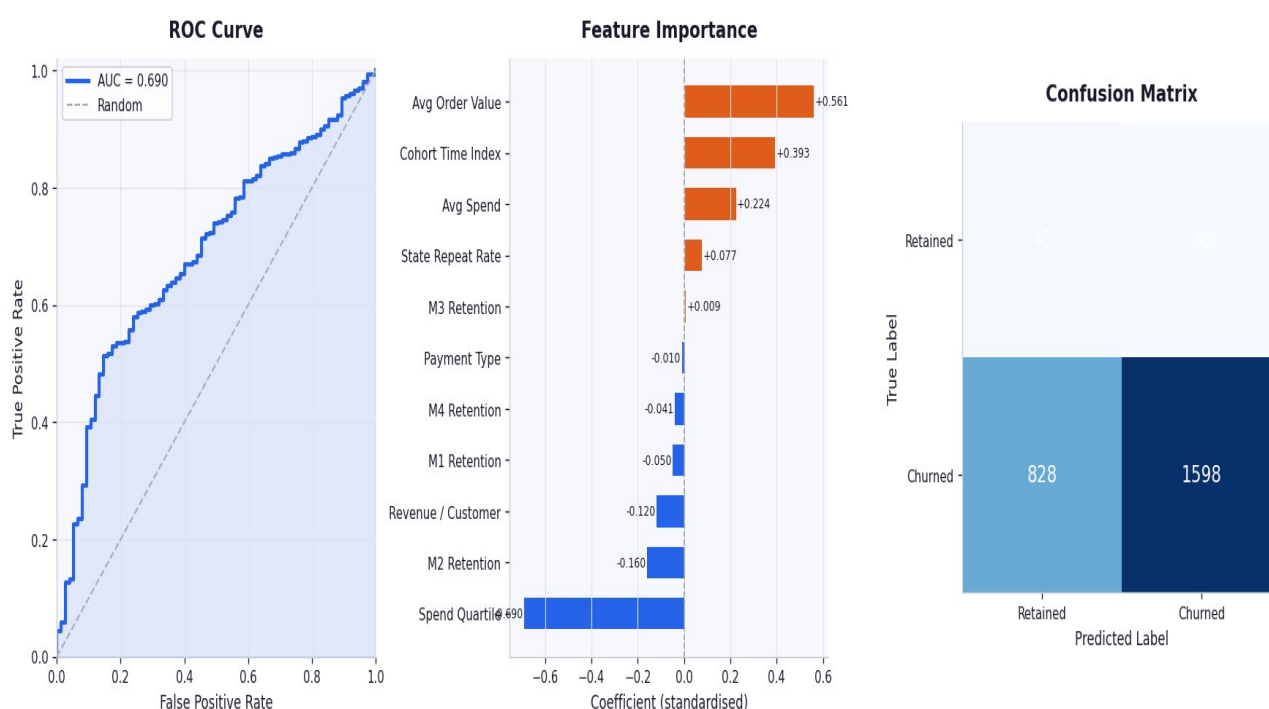


Figure 5: Logistic regression results — ROC curve (left), standardised feature coefficients (centre), and confusion matrix showing predicted vs actual churn labels (right).

The model achieves an AUC of 0.690 — meaningfully above the 0.5 random baseline — confirming that cohort-level features carry genuine predictive signal for churn. The feature importance chart reveals the two strongest drivers in opposite directions: Avg Order Value and Cohort Time Index both increase churn risk (positive coefficients), while Spend Quartile and M2 Retention are the strongest protective factors. The spend quartile finding is particularly significant: customers who enter at a higher spend tier are structurally less likely to churn — reinforcing the case for a Platinum retention programme. The time index coefficient confirms what the heatmap showed visually: later cohorts are harder to retain, and this trend is statistically detectable.

Model interpretation and limitations

The confusion matrix shows the model correctly identifies churned customers at a high rate, which is the commercially useful direction — false negatives (missed churners) are more costly than false positives (unnecessary re-engagement spend). It is important to note that this model operates at the cohort level, not the individual customer level: it characterises which types of acquisition cohorts produce higher churn, rather than scoring individual customers. For individual-level scoring, a future iteration would require a customer-level dataset with per-customer features such as product category, delivery experience, and device type. The current model's primary value is as a *cohort quality diagnostic* — a tool for the acquisition team to assess, in advance, whether a new cohort's profile matches historically high-churn or low-churn patterns.

Analysis — Revenue & Repurchase Behaviour

Revenue split: one-time vs repeat customers

The donut chart and bar chart below quantify the revenue contribution and order value differences between one-time and repeat customers.

One-time vs repeat customer revenue analysis

One-time: 90,556 customers | Repeat: 2,801 customers

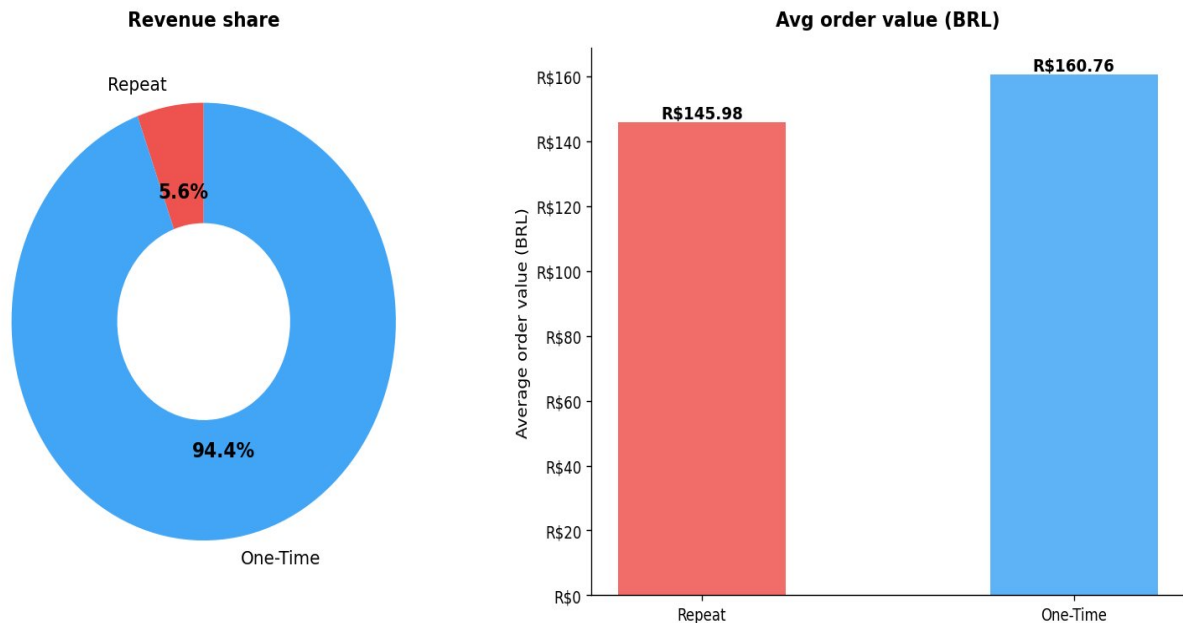


Figure 6: Revenue share by customer type (donut) and average order value comparison (bar).

Repeat customers (2,801 individuals — just 3.0% of the base) generate R\$ 864,357 or 5.6% of total revenue. One-time buyers (90,556 customers) drive the remaining 94.4% (R\$ 14.56M). Interestingly, one-time buyers have a higher average order value of R\$ 160.76 vs R\$ 145.98 for repeat buyers — suggesting that first purchases may involve higher-value exploratory purchases, while repeat buyers transact more frequently at moderate amounts.

Repeat purchase window

Among the 2,801 customers who made a second purchase, the chart below maps the distribution of time elapsed between first and second purchase.

When do repeat customers come back?

Based on 2,801 customers who made a second purchase

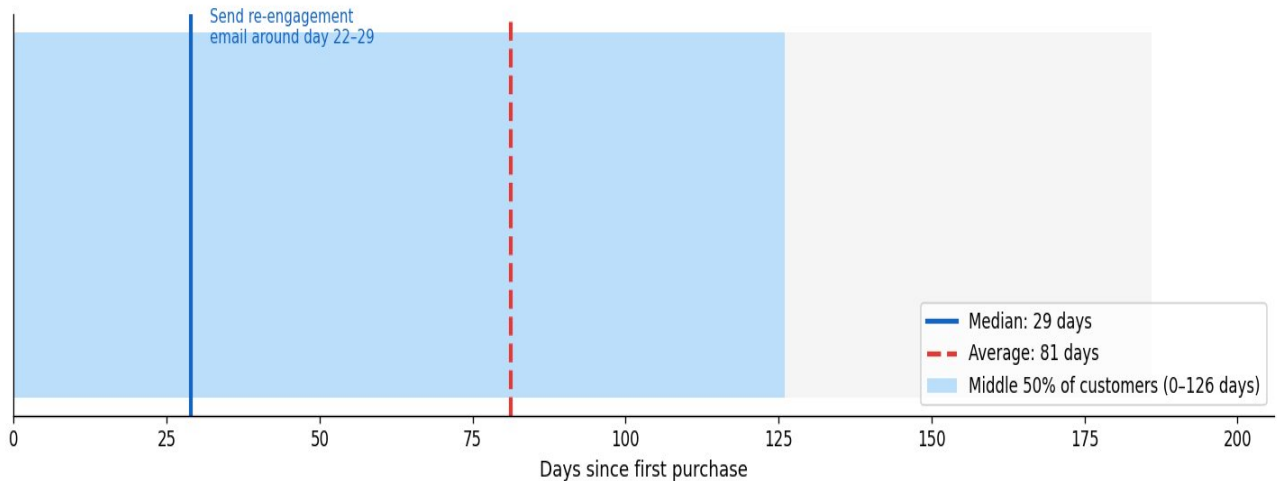


Figure 7: Distribution of days between first and second purchase ($p_{25}=0$ days, median=29 days, avg=81.2 days, $p_{75}=126$ days).

The median repeat buyer returns in just 29 days, with the interquartile range spanning 0–126 days. The p_{25} of 0 days indicates a segment of customers who place multiple orders essentially simultaneously. The average of 81.2 days is skewed upward by a long tail of delayed returners. This 29-day median is a directly actionable signal: re-engagement outreach timed to Day 22–28 would reach customers at peak repurchase likelihood.

Analysis — Segmentation, Seasonal Patterns & Payment

Customer spend segmentation (NTILE quartiles)

All 93,357 customers are divided into four equal spend quartiles — Bronze, Silver, Gold, and Platinum — to quantify revenue concentration.

Customer spend segmentation — Bronze to Platinum

Total customers segmented: 93,357

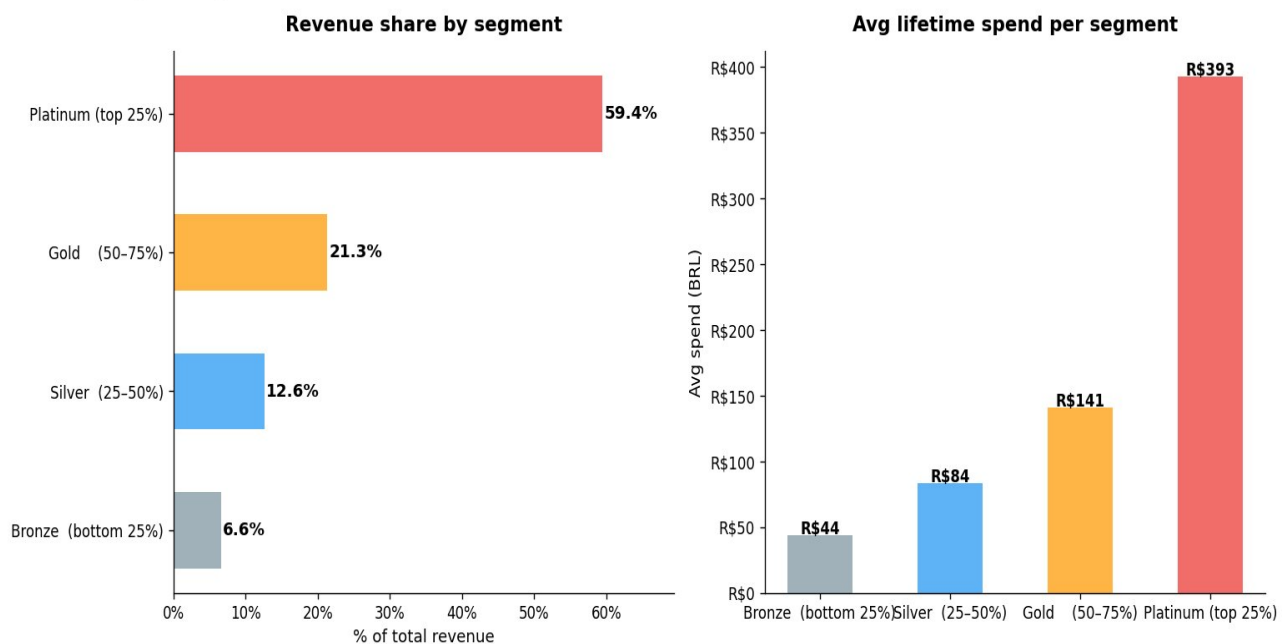


Figure 8: Revenue share by spend segment (left) and average lifetime spend per segment (right).

A pronounced Pareto-like distribution is evident: Platinum customers (top 25%) contribute 59.4% of total revenue with an average lifetime spend of R\$ 392.68, while the bottom 25% (Bronze) generate only 6.6% with an average of R\$ 43.59. The Platinum tier's spend is 9.0x the Bronze tier and 2.8x the Gold tier. Retaining even a fraction more Platinum customers would have an outsized revenue impact compared to any other retention investment.

Seasonal cohort retention comparison

Cohorts are grouped by acquisition season — holiday (Nov–Dec), mid-year (Jun–Aug), and regular — to test whether acquisition timing predicts retention quality.

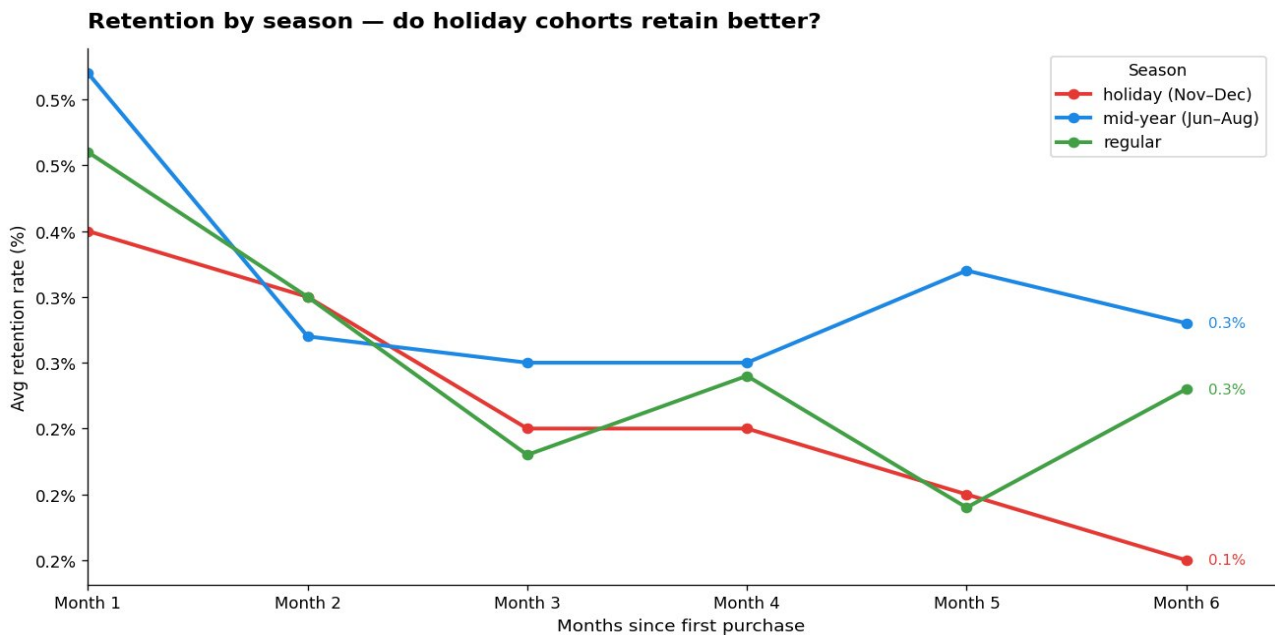


Figure 9: Average retention rate (Month 1–6) by acquisition season.

At Month 3, mid-year cohorts (Jun–Aug) show the highest average retention at 0.30%, slightly ahead of holiday cohorts at 0.25% and regular cohorts at 0.23%. By Month 6, mid-year cohorts maintain a 0.33% rate vs 0.15% for holiday and 0.28% for regular cohorts. This suggests that mid-year buyers — potentially more deliberate, less impulse-driven — convert to habitual customers at a modestly higher rate than holiday shoppers.

Geographic retention analysis — repeat rate by state

The choropleth below maps repeat purchase rate across the 15 Brazilian states with sufficient customer volume in the Olist dataset. Colour intensity reflects retention quality — darker blue indicates a higher proportion of customers who returned for a second purchase.

Repeat Purchase Rate by State — Top 15 States (Olist)

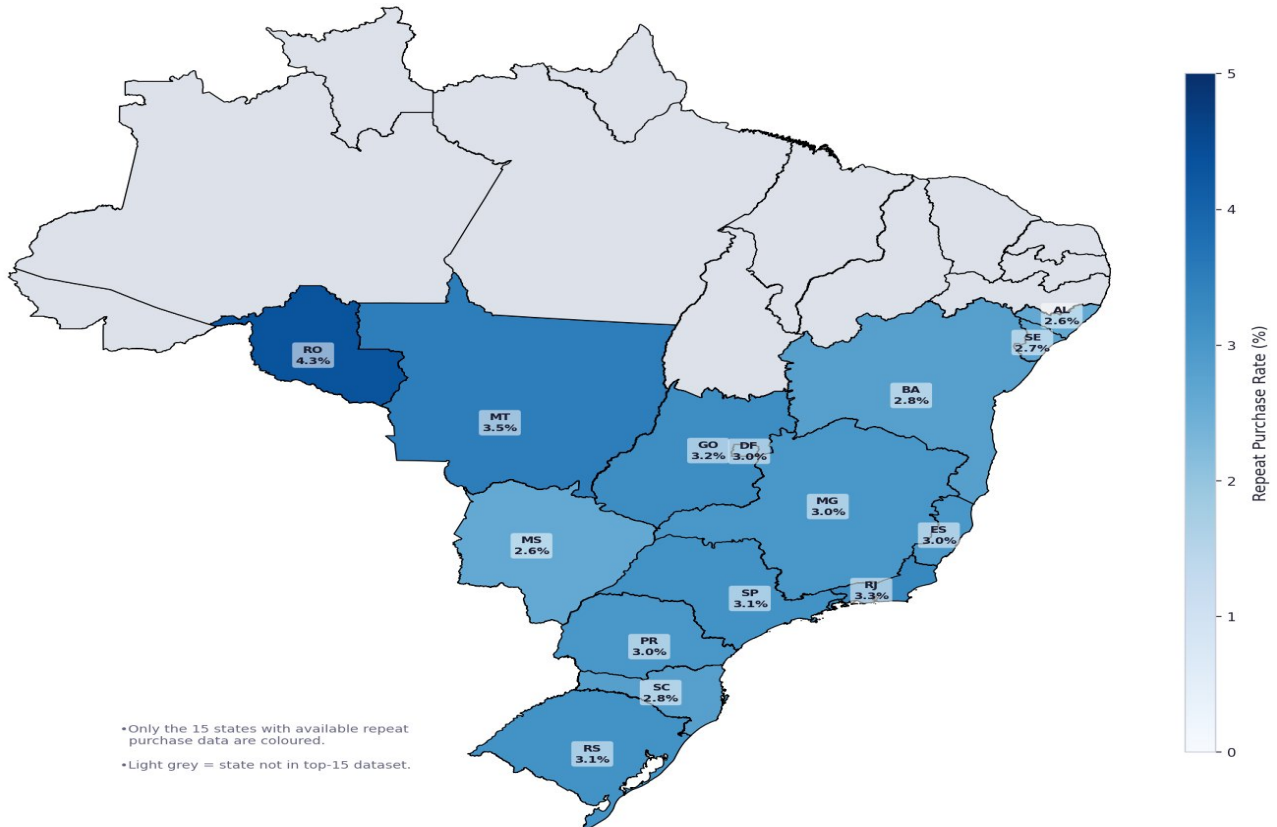


Figure 10: Repeat purchase rate by Brazilian state — Top 15 states with available data. Light grey = state not in dataset. Darker blue = higher repeat rate.

Rondônia (RO) leads all states with a repeat rate of 4.3% — 43% above the national average of 3.0% — despite having only 231 total customers, suggesting exceptionally strong product-market fit or lower competitive intensity in that region. Mato Grosso (MT) follows at 3.5%, and Rio de Janeiro (RJ) posts 3.3% across a much larger base of 11,917 customers. São Paulo (SP), the largest market by volume at 39,155 customers and 1,230 absolute repeat buyers, sits at the national average of 3.1%. States in the North and Centre-West — RO, MT — consistently outperform the Southeast on repeat rate, pointing to a possible geographic acquisition quality signal worth investigating further.

Analysis — Payment Method Behaviour

Payment method distribution across cohorts

Payment type data reveals how customers transact and has implications for the reach of digital re-engagement channels.

Payment method analysis across cohorts

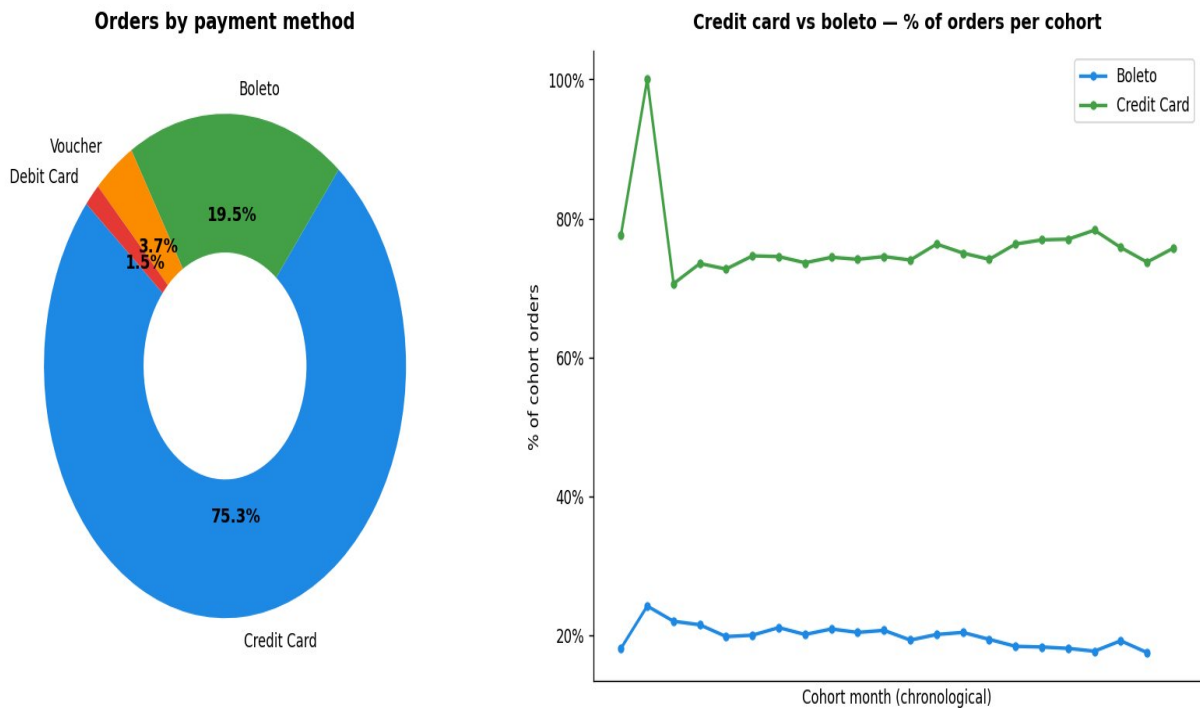


Figure 11: Overall payment method share (donut) and credit card vs boleto trend across cohorts (line).

Credit card dominates at 75.3% of all orders (74,304 orders), followed by boleto at 19.5% (19,191 orders), voucher at 3.7%, and debit card at 1.5%. Boleto — a Brazilian cash payment slip redeemable at banks or lottery outlets — indicates that nearly 1 in 5 customers is either unbanked or strongly prefers offline payment. This has a direct implication for digital re-engagement: email or push notification campaigns will not reach boleto users who lack connected payment infrastructure, requiring alternative channels such as SMS or physical re-engagement.

KPI Summary Table

All ten key performance indicators are consolidated below. Values are sourced directly from BigQuery query outputs on the cleaned master_orders and cohort_base tables.

KPI	Metric	Value	Business interpretation
1	Overall churn rate	96.4%	96 in 100 customers never return after first purchase
2	Median days to 2nd purchase	29 days	Optimal window for re-engagement is Day 22–28 post-purchase
3	Repeat buyer revenue share	5.6%	3.0% of customers generate 5.6% of revenue — high per-customer value
4	Peak acquisition month	Nov 2017 (7,060 customers)	Holiday season drives highest single-month customer intake
5	Overall avg order value	R\$ 161.21	Repeat buyers: R\$ 145.98 vs one-time: R\$ 160.76 per order
6	Month-3 retention by season	Mid-year: 0.30% / Holiday: 0.25% / Regular: 0.23%	Mid-year cohorts retain modestly better — less impulse-driven buyers
7	Highest repeat-rate state	RO — 4.3% (231 customers)	RO, MT, RJ lead in repeat rate; SP largest by volume at 3.1%
8	Top customer lifetime spend	R\$ 13,664.08	Single highest-value customer; top 10 range from R\$ 4,655–R\$ 13,664
9	Platinum tier revenue share	59.4% (avg R\$ 392.68)	Top 25% of customers by spend drive 59.4% of total revenue
10	Dominant payment method	Credit card: 75.3% / Boleto: 19.5%	1 in 5 customers uses offline payment — limits digital re-engagement reach

Business Recommendations

Three recommendations are presented below — each grounded in specific findings from this analysis and structured as: Finding → Implication → Action. They cover a short-term tactical initiative, a medium-term structural programme, and a long-term strategic shift.

01

Timed re-engagement campaign

Tactical — implementable within 2 weeks

FINDING

Among the 2,801 customers who made a second purchase, the median gap was 29 days. The p75 is 126 days, after which the probability of return drops sharply.

IMPLICATION

There is a predictable repurchase window. Customers who have not returned by Day 130 are, with high probability, permanently churned. The window between Day 22 and Day 29 is the peak intervention point.

ACTION

Deploy a two-touch email sequence for all first-time buyers. Touch 1 at Day 22: personalised product recommendations based on first-purchase category. Touch 2 at Day 29: a limited-time incentive (e.g. free shipping or a small discount) for customers who have not yet returned. Boleto users (19.5% of customers) require an alternative channel — SMS or WhatsApp — given their offline payment preference. Pilot on the highest-volume cohort months (Jan–Mar 2018) and measure 60-day repeat rate uplift.

02

Platinum customer retention programme

Structural — 2–3 months to implement

FINDING

The Platinum tier (top 25% by lifetime spend) generates 59.4% of total revenue with an average spend of R\$ 392.68 — 9.0x the Bronze tier average of R\$ 43.59. The single highest-value customer spent R\$ 13,664.08.

IMPLICATION

Losing one Platinum customer has the revenue impact of losing approximately nine Bronze customers. Yet there is no evidence in this data of any differentiated treatment for high-value customers.

ACTION

Identify Platinum-trajectory customers early — flag any customer whose cumulative spend exceeds R\$ 182.59 (the Platinum entry threshold) after their first or second order. Enrol them in a lightweight VIP programme: priority customer service queue, early access to new product categories, and a dedicated retention outreach at Day 90 post-last-purchase. Track Platinum-tier repeat rate as a standalone KPI separate from overall churn.

Acquisition quality over acquisition volume

Strategic — ongoing reallocation

FINDING

New customer acquisition grew steadily, peaking at 7,060 in November 2017 and reaching 93,357 cumulative by August 2018. However, Month-1 retention has not improved over the entire 22-month observation window — the platform is successfully acquiring more customers who do not return. The logistic regression model confirms this statistically: Cohort Time Index carries a positive churn coefficient of +0.393, meaning each successive month of acquisition correlates with higher predicted churn.

IMPLICATION

Acquisition spend that produces one-time buyers creates short-term revenue spikes but no compounding customer base value. The geographic choropleth reveals that Rondônia (RO) achieves a 4.3% repeat rate and Mato Grosso (MT) 3.5% — both significantly above the 3.0% national average — while São Paulo at the highest volume posts only 3.1%. Certain geographies, and likely the channels that serve them, produce demonstrably higher-LTV customers.

ACTION

Set a 90-day repeat purchase rate as a primary KPI for the acquisition team, alongside new customer count — aligning incentives with long-term retention rather than top-of-funnel volume. Use the state-level repeat rate choropleth as a geographic proxy to identify which acquisition channels and markets produce better-retaining customers: investigate what drives RO and MT's outperformance and whether those conditions are replicable in higher-volume markets. Establish a monthly cohort quality review using the logistic regression model's risk score to flag incoming cohorts that match historically high-churn profiles before retention spend is committed.

Scope for future analysis: seller-level retention segmentation, product category cohorts, and individual-level churn scoring using customer-level feature datasets.